SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA1603
COURSE NAME: Data Warehousing and Data Mining
NAME:P.GAYATHRI
REG.NO:192425277

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding *(2)*	Method Selection *(2)*	Solution Process *(2.5)*	Accuracy of Calculation *(2)*	Interpretation of Results *(1.5)*	*Total(10)*
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _P.GAYATHRI/192425277_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P.GAYATHRI

25-7-26
Saturday

Activity - 4

CO2 - AT - 1

1) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

a) What is the mean of the data? What is the median?

b) What is the mode of the data? Comment on the data modality (i.e. bimodal, trimodal, etc.)

c) What is the midrange of the data?

d) Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?

e) Give the five-number summary of the data.

f) Show a boxplot of the data.

i) Mean:

$$\frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\frac{13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+45+46+52+70}{27}$$

$$= \frac{809}{27}$$

$$= 29.96$$

ii) Median

13+15+16+16+19+20+20+21+22+22+25+25+25+25+30+33+33+35+35+35+35+36+40+45+46+52+70

$$= \frac{27+1}{2} = \frac{28}{2} = 14$$

Median = 25

2) Mode

16	2
20	2
22	2
25	4
33	2
35	4

mode are 25 & 35 (bimodal)
As they are only two were repeating it is bimodal.

3) midrange

$$\text{midrange} = \frac{\text{Min} + \text{Max}}{2}$$

$$= \frac{13+70}{2} = \frac{83}{2} = 41.5$$

4) First Quartile

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25

$$\frac{13+1}{2} = 7$$

$Q_1 = 20$

upper half

30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

= 7

$Q_3 = 35$

5) Five no summary

min = 13

$Q_1 = 20$

median = 25

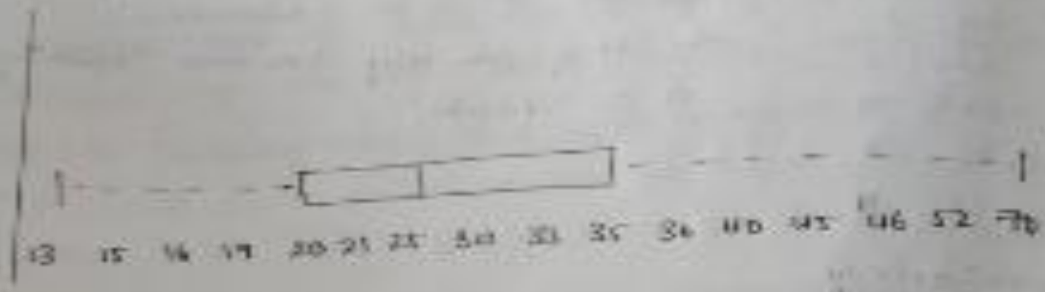
$Q_3 = 35$

max = 70

$\boxed{13, 20, 25, 35, 70}$

6) Boxplot

Boxplot



(9) Given the following measurements for the variable age: 18. 22, 25, 42, 28, 43, 33, 25, 53, 28, standardize the variable by the following:
compute the mean absolute deviation of age.

Ans Mean = $\frac{330}{10}$

$= 33$

Mean = 33 x(age)	$\lvert x-\bar{x} \rvert$
19	15
22	11
25	8
42	7
28	5
43	10
33	9
31	2
56	23
28	5
	88

$\boxed{MAD} = \frac{\sum \lvert x-\bar{x} \rvert}{n} = \frac{88}{10} = 8.8$

(3) Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. Use smoothing by bin means to smooth the above data using a bin depth of 3. Illustrate your steps.

Ans)
$$\frac{13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70}{27}$$

$$= \frac{809}{27} \quad 29.96 \approx 30$$

bin depth = 3

$$\text{Number of bin} = \frac{\text{Dimension}}{\text{bin depth}} = \frac{27}{3} = 9$$

Bin	values
B_1	13, 15, 16
B_2	16, 19, 20
B_3	20, 21, 22
B_4	22, 25, 25
B_5	25, 25, 30
B_6	33, 33, 35
B_7	35, 35, 35
B_8	36, 40, 45
B_9	46, 52, 70

Compute mean		Mean	smoothed value
B_1	13, 15, 16	14.67	14.67, 14.67, 14.67
B_2	16, 19, 20	18.33	18.33, 18.33, 18.33
B_3	20, 21, 22	21	21, 21, 21
B_4	22, 25, 25	24	24, 24, 24
B_5	25, 25, 30	26.67	26.67, 26.67, 26.67

B1	33, 33, 35	22.67	22.67, 22.67, 22.67
B2	34, 34, 35	35	35, 35, 35
B3	36, 40, 45	40.33	40.33, 40.33, 40.33
B4	46, 52, 70	56	56, 56, 56

(4) Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalization

$$x' = \frac{x - min(x)}{max(x) - min(x)} \Rightarrow x' = \frac{x - min(200)}{max(1000) - min(200)}$$

min = 200
max = 1000 $\quad \frac{x - 200}{800}$

a)

Original value	Calculation	Normalized v
200	200-200/800	0
300	300-200/800	0.125
400	400-200/800	0.250
600	600-200/800	0.500
1000	1000-200/800	1

Min-max normalized data

0, 0.125, 0.250, 0.500, 1

b) Z - score Normalization

$$\mu = \frac{200+300+400+600+1000}{5} = \frac{2500}{5} = 500$$

Mean = 500

x	$x-\mu$	$(x-\mu)^2$
200	-300	90000
300	-200	40000
400	-100	10000
600	100	10000
1000	500	250000

$\Sigma (x-\mu)^2 = 90000 + 40000 + 20000 + 250000 = 400000$

Variance

$$\sigma^2 = \frac{400000}{5} = 80000$$

Standard deviation

$$\sigma = \sqrt{80000} \approx 282.84$$

Value	z-score
200	$-300/282.84 = -1.06$
300	$-200/282.84 = -0.71$
400	$-100/282.84 = -0.35$
600	$100/282.84 = 0.35$
1000	$500/282.84 = 1.77$

Z-score : $-1.06, -0.71, -0.35, 0.35, 1.77$

(5) Suppose a group of 12 sales price records has been sorted as follows : 5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215 partition them into three bins by each of the following methods (a) equal-frequency partitioning (b) equal-width partitioning (c) clustering

No. of record = 12

No. of bin = 3

a) $\frac{12}{3} = 4$

Bin	values
B_1	5, 10, 11, 13
B_2	15, 35, 50, 55
B_3	72, 92, 204, 215

b) equal-width partitioning

$$\frac{Max - min}{No.\,bin} = \frac{215 - 5}{3} = 70$$

B_1 : 5 - 75

B_2 : 75 - 145

B_3 : 145 - 215

Assign

Bin	interval	values
B_1	5 - 75	5, 10, 11, 13, 15, 35, 50, 55, 72
B_2	75 - 145	92
B_3	145 - 215	204, 215

c) clustering

cluster	value
C_1	5, 10, 11, 13, 15
C_2	35, 50, 55, 72, 92
C_3	204, 215

(6) Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the following result.

age	23	23	27	27	39	41	47	49	50
% fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
% fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

a) Mean, median and standard deviation.
b) Draw the boxplots for age and % fat
c) Draw scatter plot and q-q plot
d) Normalize the two variables based on z-score
e) calculate the correlation coefficient

a)		Age	% fat
	Mean	46.44	28.78
	median	51.09	30.70
	Standard	13.22	9.25

(b) Box plots

min, Q_1, median, Q_3, max.

23 27 39 41 47 49 50

(c) Scatter plot and Q-Q plot

x-axis - Age
y-axis - % fat

Observation:

* The scatter plot shows an upward trend
* As age increases, body fat generally increases

Q-Q plot

Draw separate Q-Q plots for

* Age
* % fat

d) z-score Normalization

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Where

μ = mean
σ = standard deviation

For age

$$z = \frac{Age - 46.44}{12.22}$$

For % fat

Age = 23

$$z = \frac{23 - 46.44}{12.22} = -1.77$$

Fat = 9.5

$$z = \frac{9.5 - 28.78}{9.25} = -2.08$$

50

2) pearson's correlation coefficient

$$r = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma(x-\bar{x})^2 \Sigma(y-\bar{y})^2}}$$

$$= \boxed{r = 0.818}$$

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)}}$$

x

y

x^2

y^2

xy

x(Age)	y(% fat)	x^2	y^2	xy
23	9.5	529	90.25	218.5
23	26.5	529	702.25	609.5
27	7.8	729	60.84	210.6
27	17.8	729	316.84	480.6
39	31.4	1521	985.96	1224.6
41	25.9	1681	670.81	1061.9
47	27.4	2209	750.76	1287.8
49	27.2	2401	739.84	1332.8
50	31.2	2500	973.44	1560.0
52	34.6	2704	1197.16	1799.2
54	42.5	2916	1806.25	2295.0
54	28.8	2916	1806.15	2295.0
56	33.4	3136	1115.56	1555.0
57	30.2	3249	915.04	1870.4
58	34.1	3364	1156.81	1721.4
58	32.9	5364	1082.41	1977.8
60	41.2	3600	1597.44	1908.2
61	35.7	3721	1276.49	2177.7

$n = 18$; $\Sigma x = 826$; $\Sigma y = 518.1$; $\Sigma x^2 = 41798$

$\Sigma y^2 = 16,564.54$; $\Sigma xy = 25,748.20$

$$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{(n\Sigma x^2 - (\Sigma x)^2)(n\Sigma y^2 - (\Sigma y)^2)}}$$

$$= \frac{18(25748.20) - 826(518.1)}{\sqrt{[18(41798) - 826^2][18(16564.54) - 518.1^2]}}$$

$r \approx 0.81$

As the pearson's coefficient value is approx 0.81 that is near to 1 which represent that both con- fitness are positively correlated

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation